DSA 8405 Probability and Stochastic Processes-20250904_165800UTC-Meeting Recording

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2h 39m 35s

OC **Odhiambo, Collins** 11:59

Hi, good evening. We're going to start shortly. So like within the next three of 5 minutes we'll be able to start.
OK, uh, morning everyone or evening?
OK, so um.
So we're about to just a recap of what we did last time and then we will start with we'll we'll try to see if we can move on.
To.
To queuing systems.
OK, so the what what we what we say is that there's a class of stochastic processes that we call Poisson process. There are different types of stochastic process. So Poisson is one of the class of.
Stochastic processes that comes naturally from counting process. OK, so counting process is with the number of counts within a specific well defined.
Inter what we call inter-arrival, but so so we we we saw that naturally we can build poison from counting process when we follow certain assumptions. We call those assumptions Chapman Kolmogorov assumptions.
So when we follow those four assumptions, we are able to naturally come up with a Poisson process.
And then we went ahead to show theoretically how we can build that process.
So stochastic is a way of looking at data. OK, so so generally stochastic is a way of looking at data. But now you ask yourself, how does that apply to?
To you, how does that have an implication to you as a data scientist? Now there there are broad ways of looking at it, but the main thing is that.
Um.
Once we know that count data can follow Poisson distribution, now we can go ahead and use what we call a Poisson regression. OK, so.
Whenever you have responses that are count, because you may find not all responses are real numbers, you find some responses or you'll be interested in predicting the number of counts and and the number of the nature of your data is

that you're dealing with counts.
Data, then you will be. You will have to use what we call a Poisson regression. So a typical linear regression that you know assumes that your response variable is a continuous variable. OK.
But now when your response is strictly count, then we'll talk of Poisson distribution. OK, we'll talk of Poisson distribution or Poisson regression. OK, and from your basic. Probability.
It's easy to see that link with the my blue board, but from your basic probability, it's easy to see that the mean and the standard deviation of Poisson.
Are they safe? So we can show that here right here.
Now the mean and the side deviation of poison are the same. Now what is the application of that?
What's the implication of that?
As far as.
As as as as what is the implication of of that as far as the data is concerned, the implication is that not all data will have, not all data will have that kind of a setting where.
The mean and the standard deviation are the same. So we talk of over dispersion. There are some data that's we have what we call over dispersion. So if you have data that's.
The nature of your data is over this much. Then you will talk of something else. We'll talk of negative binomial.
OK.
But just let me just do a recap shortly and then I can move on to the queuing system. So I'll I'll avoid the much abstract mathematics because I know you are a data scientist, but it's it's also important to to know to to a certain extent, especially for those of you who will be coming back to do.
Your PhD or wherever you will go to do your PhD if you desire to move forward with your education so that you passed your masters, you want to do your PhD and and then during your PhD you'll be required to.
Prove some of these concepts, so it will be important for you to have some background around it.
But for those of you who are just here professionally, you want to your work is just you want to remain. Your data science is Masters is your final degree and you just want to focus on the industry and stay there.

OK, you may not need the theoretical Bureau of it, but the theoretical Bureau of it helps in understanding some of the interpretations that you may face in the industry. OK. So, so that's why we have to balance both sides. There are, there are those who will eventually do their PhD and so you'll be needed, you'll be required to have some basic background in some of the theoretical.
Pieces of the stochastic process.
OK. So just a recap, we we looked at Poison process, so where you have within each time period there are events that occur, so events that occur over time.
So as events occur over time, then the the, the, the, the, the, the process PS is the sum. The sum of the events given time greater than T is defined as this where NT is the natural number, so the name.
The NT is a natural number, is a count. It's not a real number. OK, so it's a natural number. It's a whole, a natural number is. So it's a whole number. You talk of a whole number. So 012 that way. So that's those are the events. I I think we did discuss this. Um.
And then this is just another way of approaching to come up with a Poisson using the Laplace transforms approach, Laplace transform approach. As I said, some of you probably will find other approaches easy.
Though I find the Kolmogorov approach much easier.
So this is just another approach using the forward recurrence using Laplace transformation. For those of you who have who have solid background in applied mathematics, this is not new to you.
So the anything to do with allow place transformation will be helpful for you if you want to up to that way.
And so you can use Laplace transformation and then end up with gamma and then transform the gamma into Poisson and then you end up with a Poisson distribution.
And there's a way you can, in theoretical statistics, there's a way you can.
You can model a normal distribution to gamma and then you can model gamma to normal. They have a relationship. We call them a family of exponential distributions.
So, so there's a so they they follow certain certain structure, theoretical structure.
OK, see, let me not go there. I don't want to get it to give you a lot of theoretical, but this is available. I'll I'll post it. We'll put it available on the.
The the class platform.
Um, um. So just to say so that I don't miss that point.
The the the the way you you if you want to look at this probability distributions.

So the probably distributions are generated from families of distributions, and the bigger family is what we call the exponential family of distribution. So there's a way theoretically using Laplace transform.
We can actually switch different distributions. We can show that normal under certain conditions can be gamma and gamma under certain conditions can be Poisson and Poisson under certain conditions. So there are those mathematical.
Flexibility that you can play around. Some of them requires when you talk of transformation, there's a Jacobian transformation.
There's a Laplace transformation, there's a Jacobian, but we most most of the probability distribution. We actually use a Jacobian transformation if you're transforming variables, if you want to transform a variable.
In in in calculus, in advanced, in advanced calculus.
You have to.
Be able to transform some variables so that you make your function much. You must be able to transform your variables so that you make your functions easier to interpret and integrate.
OK.
I don't know what what how they teach nowadays. I don't know how they teach nowadays calculus, but during our years those are quite quite quite some times back. So we used to have differential calculus.
Just a we differentiate the calculus of differentiation.
So no, no, no. First you start with theory of limits. You start with the concept of limits. OK, so so you have to be very thorough in the theory theory of limits. Once you know you've cracked the theory of limits, then you upgrade, you go to differential calculus.
So within differential calculus, there are techniques you play around with differential calculus. So once you're good in differential calculus, then you upgrade. You go to the integral calculus now.
Not all functions. Most functions are differentiable, but not all functions are integral. So there are there are there are some difficulties you find with certain functions when it comes to integration.
Asian.
OK. So when it comes to integrations, we face some difficulties and once you when you face those difficulties, then there are techniques we deploy in calculus. Some of them are a bit advanced.

Some of them are are not very advanced, but there are certain techniques to deploy in curriculars and and and and depending on the functions that you're looking at, if you want to integrate, you may find difficulties if you use a normal.
Integration, the power rule that most of you know. So there are certain functions that you may find difficult and that require to deploy some more advanced techniques.
So one of those techniques is transformation. So when you do transformation then you need to.
Know there are some certain advanced techniques of transformation, like Jacobian or Laplace transformation. OK, so by then you also have to be a little bit knowledgeable on differential equations, a bit of.
Differential equations and then we have difference, difference equations. I used to teach calculus many years ago before I completely went just delve into statistics and research.
But it's that there are certain concepts that are a bit advanced in applied mathematics, but to those concepts, we still need them in probability theory. OK, so I'll not take you to.
Those advanced areas, but I'm just sharing with you because when it comes to probability distribution.
Uh, when it comes to probability distribution.
If you want to understand probability distribution, you have to understand exponential distribution.
OK.
So in fact, here in the United States, some of the courses I was teaching in senior courses I was teaching at at University of California, some of the advanced courses I was teaching were.
On advanced probability, I had to spend a bit of time to take students through calculus first before we can go to apply them in probability theory. So I will take like almost two weeks and just focus on.
Calculus, because some of them have no idea, did have a lot of ideas on calculus. They just brushed through. But now when you start to do dealing with concrete probability and you start.
Interpreting concrete probability distributions, you will need to borrow a lot of facilities or a lot of a lot of techniques from calculus for you to play around with.
With us. So they they usually the best. The best approach is usually to start with exponential distribution. So we start with what is Maclaurin seeds? What is Taylor

seeds? What's what's Maclaurin seeds? OK, so this is Taylor seeds and this is Maclaurin seeds. Then what are the different forms of Maclaurin seeds?
Where does the concept of the limits comes in? So once we grasp exponential distribution then we start giving different versions of exponential distribution. So and that can take you to normal distribution can take you to theoretically.
All other types of distribution, Weibull, Pareto, log normal, log logistic, all those are forms of what we call exponential family of distributions and that includes Poisson that includes so the so the so there's.
The way you can always like I can start with normal, but under certain conditions I can play around with it to end up with exponential. I can play around with it and end up with Poisson. OK, so.
So those are different versions that I'm I was trying to make sure that you you can recall so.
So if you read different books, different people have different approaches. But this guy here is using Laplace transformation, which you may find easier or not depending of your on your orientation, your mathematical orientation.
Yeah. So of course this is now the exponential distribution. We will revisit exponential distribution when we'll be looking at queuing systems. So this is not a probability cost. So I'll I'll just jump that. I'll only pick what I think is.
Relevant for the course, but from from this concept you can you can build exponential distribution from scratch. OK, so from scratch. So here is a form is a right is a way of building a distribution from scratch continue end up with the.
Distribution we always know. Some of us what we were taught, we were just told this is exponential. It is written this way. This is Poisson. It's written this way. But if you dig down and start building it from scratch.
OK, this is how you reason in order to build it from scratch. OK, so you reason within interval, a small interval H, there are certain arrivals and then of course using the.
Are you using other approaches? OK, of course you'll end up with the exponential distribution.
But exponential distribution will be helpful for us when we'll be doing when be looking at the queuing system.
Right. So let me not to go there. I'm just summarizing the concept of spacing. The order statistics is a bit abstract, but it's very important when we are looking at modelling.
Stochastic processes from scratch, what we call order statistics.

OK. OK. Let me jump that because of its abstract form. If if you guys were doing MSE statistics and we had a course called Advanced Statistics, so I'll stochastic processes, then I'll probably.
Be happy to go take you through the these ones, but you guys are better sense, so you really don't need the most technical bit of it, but just an overview of what it is.
Yeah, so these are partial derivatives just to work on that. So I don't think you need.
And then there's a Jacobian, the Jacobian transformation. When you're transforming variables, you can approach it using Jacobian transformation.
Or what we call a Jacobian matrix or the Laplace transformation.
OK. And and do those some more, the more ordinary differential equations, the ODE, some of you have a background in mathematics, applied mathematics. I think you must have done ODE in either your third year or fourth year ordinary differential equation.
or partial differential equations.
So these things may not be new to you if you have a background in applied mathematics.
All right, so let me jump all these guys. They'll just make your life a bit tough. So I don't want to make your life tough with these guys called Jacobian. I just wanted to give us an overview of that.
And then I can jump, so I can jump here counting processes. We've already discussed counting processes and these are the Chapman Kolmogorov.
But part of Chapman called Mogorov the postulates. And then of course we can use.
You can use it to generate stationary events processes and then end up with Poisson processes.
Uh, OK, so.
I need to just move out of this.
And introduce to us, uh, the queuing systems.
Unless we have any question on poison.
Yeah.
OK, so I just need to start off with the queuing systems. The queuing system is also an extension of.
Is an extension of Poison, but now trying to incorporate certain things and I think that's what we are going to start off today.
Uh, looks like I disappeared a bit.

All right. Can you confirm somebody just confirm if you can hear me?
I think I disappeared a bit. OK, OK, OK.

Elias Shamalla 41:48
Yes, yes, we can hear you now.

Odhiambo, Collins 41:53
Thank you.
Uh again.
The queuing system is also another branch of stochastic processes that is with queue.
So we look at how the queuing component is made-up. What's what? What do we mean by theory, queuing theory and the structures of a queue?
Some and some of the assumptions when we are modelling AQ and then we look at the mathematical setting and then we'll be able to model a very simple queue. I I should say also that.
When when it comes to modelling, you we usually start with very strict assumptions, but we depending on reality on the ground, you adjust to those assumptions. And of course the more you adjust, the more complicated the mathematics is.
Nowadays there's a lot of advancement in a I and data, so that you may be you may be able to navigate those difficulties easily, but it's always good to have the concept, the the, the mathematical, the theoretical concept of it.
So that now you can adjust appropriately depending on the situation you're dealing with. So for example, when we were looking at the Poisson process, we had these four assumptions, which we are calling Chakman Kolmogorov assumptions.
Now the reality is that you may find probably some of those assumptions are not practical, so you may have to adjust some of them, but now introduce some mathematical components that take care of the adjustment.
And that's how we do modelling. So when you what, what, what we mean when when you talk of modelling, you start with very strict assumptions, but you go on reducing and becoming more real, more realistic. So in this class we will model very simple, the easiest cheap.
Lowest, simplest, whatever kind of model. So just to give you an idea of how modeling is done, stochastic modeling is done and and we will look at the queuing system and how that.
Uh, works.

So one class of stochastic models are the model associated with the queuing systems or what we call queue that results when the demand of a process exceeds the current capacity of that process. It's like you when you walk into a bank.
And want to either withdraw or deposit or do some transactions within the bank, you will naturally form a queue if you find that there are more customers there. So the service, if the demand exceeds the current capacity of the process, then a queue will naturally.
Be formed. So that's what we mean by queuing system, just from the word queuing. OK, if you go to the cafeteria at Strathmore and you find people are lining up, it means that there are more people and that those more people the demand is high compared to.
The rate at which service is done. So if that exists, then you naturally form what we call a queue.
So a queuing system consists of one or more servers. So somebody has to serve. There has to be service mechanism that provides service of some sort to arriving customers. So customers arrive and then they arrive.
And they have to queue in order to receive service. So customers who arrive to find all servers busy generally join one or more of the queue line in front of the server. Hence the name we call queuing systems just from the.
From that basic layman description of of what we mean by a queuing system.
So by from that layman description of what we mean a queuing system is that to any walk to take service or to receive service, you will naturally join a queue as long as the rate of service is very high.
Is is lower, is lower. The service mechanism takes a bit more time compared to the rates at which customers arrive, arrive into into into this for service for specific service.
So customers who arrive find all servers busy generally join one or more queues. Example of queuing system include a bank teller, computer system, computer system. So when you have a.
Let's say a server.
You you have a main server but different terminals are sending queries. So that one we talk about computer system in ICT. OK, so so think of it like think of it like Nairobi Hospital.
Nairobi or Grakan so so in Nairobi hospital.
When when patients arrive, OK at the triage, so so your patients arrive at the triage.

They will naturally form a queue. OK, so as long as there is nobody is being served, so the rate at which the service mechanism is taking more time compared to arrival of patients. OK, so that's one type of queue, but there's another type of queue in computer system.
Because at at the at the triage, at at the at the reception desk there, the person dealing with accounting will check your either a medical card or if you're paying by cash.
But we send query to the central server. So you send queries to like pull out your information, your patient information. So the query will go and search from the main server and then are they triage the nurse who is speaking oral?
Collecting variables like your blood pressure, the systolic, diabolic measures and and all other measures, heart rate, blood pressure, weight, whatever.
That.
Clinician is at the triage is also sending queries to the main cell. Now from that point you go to see the doctor. The doctor will also send queries to the main cell.
OK, so I'm assuming it is a paperless environment and then the doctor will probably send the patient to the lab or to the pharmacist. So patients who are in the lab are also the the the lab technician.
Is also sending or she's also sending queries to the main server and then the pharmacist is also doing the same or the radiologist. If they feel you need, they need to do some radiology, they also do the same. OK, so so we also have.
Computer systems that operate in a queuing system kind of application. So I wanted to see that when we talk of queuing system, it doesn't necessarily mean people, it can mean entities, it can mean you can look at the computer environment, you can look at the manufacturing system.
Systems OK where a raw material arrive and then service mechanism is to process that raw material. You can look at the maintenance of the system, you can look at communication system.
Like if you call Safaricom and you have issues with your M-pesa or your call or whatever your line, you have issues with your line you want to be solved. If you call Safaricom and it goes to the customer care desk.
That is a queuing system because you'll be put on the queue for a while. OK, so the the customer service person is able to make respond to calls. So that that response to call is what we call service mechanism.
But the fact that you have arrived, you have called, that's an arrival.

So the the the way it hits the the customer service telephone ball, telephone whatever booth that that's an arrival. So that's that's actually like a poison process like the the the calls, the number of calls have arrived.
In a certain time, inter arrival time. So that's an arrival, but once the call arrive then they are put on a queue so that now the customer, the customer service person representative will respond to you after they have finished with the people who are on the queue.
So that's also a queuing system can be modelled there air traffic control.
At Kenya Airport Authority, KAA does this.
In air traffic control. So arrivals are the aeroplanes that arrive in an airport.
And then service mechanism, there are certain engineering aspects that they look before they are. So the aeroplane people will, will, people will disembark, then there will be a bit of service done to the aeroplane before another.
Uh, before it takes off. So that's service mechanism.
So the principal actors in a queuing situation are the customers and the servers. So these actors meet to form a queuing system.
So that's that's a general view of our giving system. It looks simple.
Um.
So looks looks simple, but we will see how the mathematics play around. Yeah.
So just trying to break it further in terms of the three components, there are three components of a queuing system.
So there's the arrival process, so where entities arrive. So you have the arrival process where entities arrive according to certain arrival pattern. Usually that's poison. So arrivals are independent of preceding arrivals that is successfully in arrival times as.
Statistically independent and completely random. So for us to model this type of stochastic process, we have to assume that the arrivals are independent and they are completely random. OK, so this one means that let's say for example.
Uh, I want to build a model like you at uh, Stan chat.
Or office at Stanchat Bank in Moya Ave. or wherever Kenyatta Ave.
I don't know if is still there. It used to be there when I was there.
Uh, so assume uh, I want to model.
The queuing system so that eventually I can advise. I can advise the manager on how to manage the queues.
OK, so I have to assume that people who arrive into the bank hall are random. So people who are arriving at the bank hall are random. They are random people

number. OK, so the number that arrives are random. So when the bank opens at 9:00 between.
9 and 10, if there were 30 people and between 10 and 11 there were 50, those numbers are independent and random. That's what we mean by inter-arrival time.
Times are statistically independent and random, so the numbers that come.
Between any inter time that I've scheduled, those numbers are random and they are independent. If it's if it is modelling, let's say I'm modelling a hospital patients that arrive at Aga Khan Hospital.
I have to arrive to assume that the patients that are arriving, they're arriving at Randall within every any specific time period, the number that are coming to the hospital Randall. So that's one of the components that we look at when we are modelling the queuing system.
The other component is the queuing discipline. So the queuing discipline represents the number of customers waiting for service, like how are they served. The queue represents that number, but the discipline represents the way the queue is organised.
OK, so you so there's a number that being solved within the queue, but now how that queue is organised that help that's helpful when modelling A queuing system.
OK, so obviously we always know the popular one is fast in, fast out. Once you walk into a bank, the first people to arrive are still the first people to be sat or you walk into a hospital in many situations. So we we talk of fast in.
Fast out, but you can have situations where you have lasting is the one that is fast out.
Where do we find that kind of a situation where you have the way the queue is organized is that the last item or entity to arrive is the one that is pushed out.
Anyone who can think of that type of practical situation?
Last in, fast out.
Bernard, what do you think of any situation like that?
Or Cosmas. Are you guys there?

Cosmas Kungu 57:15
Yes, we are there. I think it is in the matter. The people who sit on the front will get out first and then the persons who arrive last week. Yeah. So I think a matter to something kind of, yes.

OC **Odhiambo, Collins** 57:29
Oh, OK.

CK **Cosmas Kungu** 57:31
That's a we had example.

OC **Odhiambo, Collins** 57:32
Yeah, any other situation?
Any other situation?

BO **Bernard Ogol** 57:39
OK, OK, I think I did this under our data structures.
We're dealing with the stock overflows.

OC **Odhiambo, Collins** 57:51
How many of you have ever worked in a warehouse?
I did. I did those days before. During our years, you used to take two years before between finishing Kenya certificate of secondary education and joining university. If you had qualified, we used to stay home for two years.
So if some of you have who have not experienced that, you're a bit lucky. So for us we used to stay by the time, by the time you finished KCSC and you joined first year in the university, you were we were home for two good years. So during those two good years.
We used, some of us, we used to go and look for a job to work at industrial area. So I did a bit of those manual job there those days, so.
So they we used to work at the warehouse. What was the company called it? I don't know if those companies are still there. They must have moved to other countries. So we used to have some of the most popular employers then Kazizam Kono.
I was called at family, was Unilever plastic. There's these guys who are doing who are making Palloro plastic. So those are some of the guys, the players we will go with, you will go to industrial area with.
Your ID and stand there and then they will get you into work for then it'll be paid per hours. We used to be paid 33 shillings per hour.
That was a lot of money then.

Yeah, so, but it's just, I I think just a joke because just just a just a joke, a joke. I think we used to be paid on Fridays those days.
And it was on cash. So you'll be paid on cash. They count the number of hours you worked for a week and then you paid on cash on Friday. It was, yeah, it used to be around Friday then. So immediately you'll be paid. You'll put your money in the socks.
And then run like because there was a lot of Bush around there between between Jogoro debt, those, those, those, those companies and there were a lot of **** along the way who.
They will know when we are paid, so I don't know who clicked to them that we have been paid. They'll know which company we had paid specifically, so they'll stand on the road.
Mode.
And those days, I don't know how past we were. Now you could not be caught. Just put your money inside your socks and disappear. But before that, those guys will organize themselves. They see you on the other side. But anyway.
Just a break. So we used to work. I used to work at a warehouse. So what we used to do is to carry products and load them into the whatever to the loading.
Uh, for dispatch and.
This concept of first last in, first out was would work because the way we would look at the batches would arrange first of all from the factory, we'll arrange the products. There are several projects, whether they are Colgate or whatever, so arrange them in batches.
So the the last batch, the the newest batch would always put them at the end so that the the last batches are the first to be dispatched. OK, so when when the logistic supervisor comes and he gives you the list.
So your work was to go pick the last batch and put them on the lorry. So probably it was delivering to Nakumat Uchumi supermarket. Then it was Uchumi was the main, the main vendor.
Or many other big shops around Nairobi. Yeah. So there was that lasting that that that was a very good example in a warehouse, especially if you're dealing with.
The items that go bad or that are perishable, like flowers or any other items or products that are perishable, you'll always deal with.
Like make sure you deal with the last inner, the ones that can go out as far so that you clear the way out. So that's just an example of.

Last in, first first out. And then there's also another discipline we call 0 served in random order.
OK, so I don't know where you can find this served in.
Where can you find a situation where you have served in random order?
Like the order which they appear doesn't matter.
Do you know of any situation like that exactly?
Or Elias. Elias, have you faced any situation where you have customers arrive, but the way they are treated, they are treated in random, Jacqueline.

Elias Shamalla 1:03:44
Uh, can we?

Jackline Mboya 1:03:44
I've not encountered that.

Elias Shamalla 1:03:50
Can we talk of, can we talk of the lottery, the gambling?

Odhiambo, Collins 1:03:51
Glass, can you think of anything? Yes.
Yes, yes, perfect. That's perfect. That's what I wanted because there's some probability modelling that is in major lottery gambling and that's what they do. It's solved it.
At Randall, that's that's perfect. That's a very good example. Um, what about priority queue?
Priority queue.
Which which practical situation can you see that can you find priority queue?

Cosmas Kungu 1:04:43
Can we speak of hospitals? Because the people who have emergencies will be given high priority and also where that the govern like airports where we have the government agencies, they queue first or the diplomats are then now there.
What's that?

Odhiambo, Collins 1:05:00

Yes.
Yes, Cosmos, that's perfect. Yes, that's perfect because yeah, so hospital you can find that. So the the reason why I'm asking you guys is just to try to provoke your mind to see that when it comes to modelling Q, it will ordinarily depend on.
The situation we are modelling because not all situations will be fast in, fast out.
So if you're dealing with data, it depends on that type of data that you're dealing with, and not all data will be faster and faster. Some will be solved at in random order.
And so people deal battling. I think there's some before I left Kenya, there was there was a company which actually approached me to do for them the mathematical model for that for sub.
In random order because they were starting to, they wanted to start a betting company, but they wanted to to have a solid mathematical probability model. So that was typically a killing system really. So that random order and so and.
So you can't talk of fast in, fast, fast out in that case or last in last out. OK, so if you're modelling arrival of patients, so you have to also consider priority queue for those who are coming.
At emergency. So, so you give priority to the most for the most needy cases at the airport. Yeah, we have business class, we have diplomats and all that, so.
Yeah, that's even Safaricom.
If Una, Una two types of customer service in Safaricom, you can call EA Manager or Kavada. You'll be put on the first in, first out, whatever for for your case to be tackled and you may wait there for some minutes.
Or an hour, but there's those line which are for premium customers, so they're given priority.

SA **Straton Amodora** 1:07:09
I think I think a good example is the banks. If you have a like huge amount, maybe the first time.

OC **Odhiambo, Collins** 1:07:18
They have a business club, is it? I see that with APSA. I think they have a business club, those business club or those with a huge amount sometimes. Yeah, they are those, they are those section where they consider those sorts.
Yeah, so, so, but for us we will focus on a very simple model for the cost sake of this

class. But I just wanted to provoke your mind to see there are dimensions, there are different varieties of queuing system.
Whereas the arrival pattern may be the same, but the queuing discipline may be different and that will determine the type of queue queuing system that you model.
And then there's the aspect of service mechanism. So service mechanism represents some activity that takes time.
And that the customers are waiting.
OK, some activities and the activities here may not necessarily be people. It could be for our CPU time slice. It could be that time slice for a CPU. When you're sending queries to a computer, a very big computer system like where I've talked of Nairobi Hospital, you may have a big server.
That house different information, but and then the terminals are attached to different terminals. OK, connecting a flight, even shooting down an enemy plane and that that that that can be a service mechanism.
And typically our service mechanism takes random time.
OK, it's not equal. Like if you go to the canteen and queue, I don't know that the person when you queue with your food, you want to pay the the the it's it's different. It differs with the time it takes to serve one.
Client compared to the app. So when we model, we look at those 3 components, then we can model like giving.
Queuing system. So queuing theory is a collection of mathematical models for various queuing systems that takes as input parameters of those three elements that we have looked at and then provide quantitative parameters to describe the performance.
Of the queue. So the whole thing is what we are calling queuing theory is a collection of mathematics. So we can put down all this down. We can put all this down mathematically and then show some some parameters and make decisions.
Around those parameters.
Because a random nature of the process involved, the queuing theory is demanding and all models are based on strong assumptions. So of course with time you would want to drop some of the assumptions so that you have a more practical queue.
Queuing terms are practically very important because of the typical trade-off between various costs, providing service and the cost associated with waiting for the service. So in terms of decision making, if you are.
Manager or you are a senior person trying to see how best you can manage your

queues. So the queuing system give us a snapshot of how best mathematically you can manage your queues. So so mathematically this is what we.
Show you this how you you can manage your queue and then we can quantify that in terms of costs. So we can even bring in the cost implication of your queue. Look at the waiting time, the average waiting time, the average number of people in the queue.
The number of people in the system and and that can help you make decisions because you can probably know which server is idle at what time.
What? What is the peak time? And do you need to hire more workers? You need to sometimes hire workers or schedule how they come to work if if probably I'm I'm the head of customer service Nairobi region at Safaricom.
I'm the head of the customer service at Nairobi Safaricom. I'll be interested to know how my workers work, my employees work in terms of their turn around time and versus.
How many people are calling at any time? Do people call mostly in the morning to to to inquire or if they have issues with their Safaricom number or their Mpesa? What's the what is?
What's the peak? OK, at what time? So I can schedule my my customer service representative so that I'm more efficient in terms of how I schedule my service representative so.
That's the world of the queuing system. So high quality fast services, yes, yes.

SK **Stephen Korir** 1:12:39
Hi, Hi. Maybe just a comment coming from the topic that you took us through. So this modeling queuing system.
Can also be connected to optimization.
Because at the end of the day, the next, the next sentence that I'm seeing there is we're trying to find a cable. So a place where we will not, the cost will not be high, will not be too high.
What would be the Tyson would be? So is it connected to the team?

OC **Odhiambo, Collins** 1:13:26
Yes, to some extent it's connected with optimization. It's just that this is just another class of stochastic process that has developed itself uniquely, but you can talk of still optimization because at the end of it.

If you are a manager, you want to use your resources effectively.
If you are a manager, you have people who report to you and you want to choose your them and deploy them effectively. OK Also you want to satisfy your customers as much as possible.
Um.
You you it's it's easy to go. I don't know if things have changed, but those those, for example, those days we used to go to to to go and look for passport before it even became worse.
We used to go and look for passport. Probably you have you have an appointment to go for passport, but you'll always meet people who are disgruntled because of the huge queue. Line is not moving. You've been there for morning. It's so huge queue, so.
And and people will always complain when they are there because it's like they are. The more you you are there, the more you waste your time. You should be doing other things somewhere, like go to work rather than just spend a whole day waiting for just to pick your passport.
Or just to submit papers, you've already done everything. You just need to submit papers so that they are verified and you leave. So managing those type of queue, you see where you you want to to make customers more.
You want to satisfy your customers and you also want to deploy your resources more effectively because if you happen to walk inside, you'll go to. It used to be called what is it Jogo House where they used to prop?
Process passports. So if you go there, you get a very huge queue, people waiting and line is not moving. People are so stressed, disgruntled, people are complaining. Nothing is happening really.
Line is not moving or it's moving at a very snail pace.
But if you walk inside where they are, you'll find other workers inside the offices who are someone just making, giving each other stories, talking, making stories. As in you even wonder, are we deploying stuff efficiently?
Probably you only need two or three people at the bucket at the office and push everyone else to come and clear and come and clear the the whatever the line. OK, so so so the queuing system is just an approach is a stochastic.
Process that you can approach and and see how best you can manage your queue and even assess the waiting time, prioritize, deploy your resources well and and and make your customers more satisfied.

Yeah, but but you can also look at it from optimization because at the end of it you want to satisfy your customers and make great use of your minimum resources that you have spent.
Spend less time. It's just that this is another class, and you will see when we break it down into mathematics, the stochastic bit of it comes in as another class of stochastic process, yeah.
OK. So so the high quality fast is expensive, but the cost cost by customer waiting in the queue are minimal. On the other hand, long queues may cost a lot because customers do not work while waiting. So that may have a very.
Pragmatic effect that is not good. So a typical problem is to find optimum system configuration and the solution is found by applying giving theory or by simulation. But for us we will focus.
On queuing theory using mathematical approaches. All right, so let's take a short break of 10 minutes, then we can come and.
Continuing with the queuing system, I think we may have to start the mathematics later or we'll start a bit, but let's take a short break of 10 minutes. So take a short break and then move on.
OK, so we need to proceed from where we stopped.
OK, so.
So there's several structures of a queue just trying to build on them.
The the the concept it see it's itself. Then we will now break it down to the mathematical presentation, but.
Although there are numerous examples of queuing system, it's possible to categorize them in one of the four.
So where you have single channel, single face, this is where entities arrive, join a single queue and is served by one server and finally departs.
OK, so where it's not complicated, you walk in, it's only one server, you join the queue, it's only one queue, and then the queue probably follows first in, first out. Then once you're served, you depart, you don't join the queue again.
What we call a single channel, single phase, but you have other queues like single channel multi phase where entity join another queue following the service for the first so you have.
Multiple services. So you join first, you join one queue, you will get served. After that you join another queue, you get served again. For example, the canteen where the first queue is to collect food.

And then the next key is to go and pay so you can have a situation where you have enjoy.
Then after finishing you join another queue. So if you're modelling that you have to consider that sequence. So we also have multi channel single face systems where an entity joins a single queue.
And is served by fast available servers. So there are several servers.
So you have one queue, but there are several servers like in a bank. So you have like 3 or 4 tellers.
The queue is 1, but now you will go to the teller who is free. So that's what we call multi channel single phase, so one queue.
But you have many different servers.
And then the other one is where you have multi channel multi phase so you have.
You have one queue. You get served by any available server. After that you join another queue.
But now you you pick the queue that is that is short and then once you arrive you are at the end of the service you join you you are picked by any server is free.
OK.
Uh, did I have? Uh, let me see if I can put that down.
Let's try to illustrate that.
OK, so we are saying.
You're saying single, single face, single channel. So you have single face and then there is a saba here.
So this is a queue. So that's that's the first, the first scenario, single phase, single channel.
OK. But we have a situation where you have single phase multiple channels, so you have one queue here.
But different.
Servers. So you have different servers here.
OK, so you have different servers, so somebody's here can be served by any of these as long as the server is free.
OK.
So you have that and then we also have the second one was so you have server and so you have a queue here.
And then you have a here.
And then after that service, then you have another queue here, then you have

another service here.
So here is that's a sub. So you form a queue, you go your sub, then you go another queue and your sub. And then there are situations where you have several multi face multi servers, so you have servers there.
So you have multiple.
He was here, so you always go to whoever is free.
So this is server 1, server 2, server 3.
And then there's also service after that. That's like in the passport.
So this could be probably they check your papers. Once your papers are OK, you go for printing. So you can either join that queue or join that queue or join that queue.
Depending on who is available, then from here you can either join that queue.
Or join this queue depending on who is available here.
So you have queues, yeah, and also you also have queues there.
OK. And then you could also have another service after that, so.
So they're so depending on which one you're looking at. So of course the the more the more it becomes like this, the more complicated the mathematical presentation.
So you have to adjust for those situations.
So for us we will only look at the simplest one, which is skew first seen first out to use this kind of skew just to give us an idea of how modelling is done under queuing systems.
So that's what I was trying to explain here. Single phase single system or you have single phase one queue multiple servers.
Multiple uh channels, but single face or multiple. So the queue will always form one of these, follow one of these for.
OK, so the study of queues determine the measure of performance of a queuing situation, including the average waiting time. So typically we'll be interested with modelling the average waiting time, average queuing length.
Among others. So this information is then used to decide on the appropriate level of service for the facility. So a decision maker may like to know the answers of some of the following questions in a queue. OK, so if you're a manager somewhere, a branch manager.
You're the head of a region and you're dealing with service to customers. You'd want to know answer some of these questions and that's what the queuing model will help us answer looking at the queue process.
So we want to know what fraction of time is each server idle. So the guys who are

serving, what fraction of time are they idle?
That will help you either reschedule how you work or change how you do business.
You want to know at what time, what fraction of time is are people are just seated and at the end of week they are paid their salaries but they are not doing anything because their work is to serve customers. But if customers are not coming then what fraction of time they are.
OK. And if they are not, if they are busy, then is it meeting the demands and all things like that? Remember at the end of this modeling is you want to be make some decisions. The idea is at the end of it.
Is for you to make some decisions. So what to know the fraction of time is a server idle? What is the expected number of customers present in the queue? What is the expected time that a server a customer spends in the queue?
What is the probability distribution of the number of customers present in the queue? And what is the appropriate distribution of customers waiting time? Among other things, you want to ask yourself some of this.
So there are several assumptions that we make when we are modelling the queuing system. So the conditions for modelling the queuing system. The first assumption is that the probably distribution on the arrival is Poisson and this one we can we had proven.
That if you have an inter-arrival time for any small change, given those Kolmogorov assumptions, we can naturally conclude that that distribution will follow up with his own distribution. So we were able to show that.
And then the service mechanism. I did not take us through the mathematics, but the service mechanism will generally follow the exponential distribution.
So the and mathematically you can prove that it's can be proven that the arrival will always naturally follow the poison and then the service, the duration it takes to serve a client will naturally follow an exponential.
PC patient.
So we also have to know the number of service channels that are there when we are doing modelling and the line discipline, which type of discipline queue the queue is following. Is it following LIFO or FIFA?
People.
And then how many people are there to service?
And the nature of that giving display.
So those are the conditions that we need for us to model a queuing system.

So there's a general notification here. This, this guy called Kendall in 1953 came up with a modification of a queue and it's what we use generally to adjust. Of course it has gone different modification over time.
But the original concept still remains, so this guy came up with a way you can classify a queue into six components.
So what he said is that you have to know the arrival pattern.
Of the items that are arriving. So we have to know the arrival, then we have to know the service pattern, the service mechanism, the duration. So you have to know the arrival, you have to know the service and then you have to know the number of servers that are there, how many people are serving.
What? What? How many service points are there? Is it one or two or whatever?
And then the killing display, is it with FIFA or anything?
OK, but usually we follow FIFA.
Capacity of the queue. The C is the capacity. Is it? What's the capacity of the of the queue?
And then P is a population size. Are you able to possibly estimate the entire population? Population can be infinite or can be finite.
OK.
Right. Looks like I'll not introduce the maths. I'll just cover the theory for today so that next week I I introduce the math, the mathematics of it.
But the basic keying model is called MM one. That's what we will model. That's what we will focus on. It's called MM one.
So this is the simplest queuing model to analyse. That's what we will do here. The arrival and the service are negative, exponential or rather Poisson. The arrival is basically Poisson and then the arrival, the service is basically.
So there's another way of calling negative exponential. Poisson is sometimes called negative exponential. In some book, those some books call it negative exponential.
So the system consists of only one server and one unlimited FIFO or queue, and then the number of customers are unlimited.
And these are the exponential. These are the assumptions.
These are the assumptions that we take when we are modeling the MM one. So we assume that the arrival is assumed to follow Poison.
The population from where the arrival originate originate is infinite, so you cannot exhaust the population from where the items are arriving. It is infinite. It's so huge like a petrol station where.

It's a highway petrol station within near a highway and it's 24 hours arrival of vehicles to be fueled.
That population of because vehicles are always passing there, that's a finite infinite population. So when they arrive at at the petrol station per hour, how many vehicles have arrived at the petrol station per hour? That becomes your arrival pattern.
And then that arrival follows poison and then service is fuel. How long it takes to fuel?
OK.
And then of course that one assumes, I don't know in in Kenya is in Kenya it may not be easy to model.
Well, like Anaya, like um.
In the US it's much easier because you fuel by yourself and in the US you just get into the petrol station and fuel and that and you go. So there's nobody to help you.
Well, you go, you scan your card, you pick there and then your card will be scanned like credit card or whatever.
And then it will run the amount. So you put to plug in the amount you want to like you. So it will run and then after that it will pick collect, pick the money from your card and then you go.
That's much easier to model. If you're looking at a petrol station, Kenya, there may be a big complication because there is there are people who.
Like somebody has to come open for you, fuel for you and probably you give them cash, they go and get change for you or whatever. So I don't know it it may, but but here it's much easier to model car that kind of a situation where you just go get the service and you go on.
Then there is no limit on the queue length, so we don't have limits on the queue length. The length of the queue can take any shape, any form.
There's no bulking from the queue, so this is another assumption we make. What we mean is that once you join the queue, you have to be serviced.
OK, you have to get the service. You have to get the service. You cannot be in the queue and in between you think this queue has taken so long, so you decide to leave without being.
Serviced. So that's what we are calling bulking. There's no bulking. Once you join the queue, that's it. You have to be served. So that's the assumption we make when we are modelling this type of queue. If if we if we adjust this assumption like include bulking, then the mathematics will be difficult, will be a bit complicated.

OK, it's doable mathematically, but now that will introduce another concept, another variable into your model where the Q is FIFO, but it there's some possibility of somebody leaving.
So for this simple queue, we assume once you join the queue, once you enter into a bank, assuming you are going to a bank for service for to to do some transactions. Once you enter into the bank and you naturally join a queue, you cannot withdraw from that queue. You have to decide.
So that's the assumptions we are making. And then the queuing mechanism is fee for fast in, fast out the the server cannot break the order or cannot decide that somebody in the middle can jump.
And go ahead of the other person. It's strictly FIFA so that so that our model works the way we are model we want to model.
And then the object one object item arrives and departs at.
At at at certain time intervals and then of course service mechanisms are seen to follow exponential distribution.
All right. So these are the mathematical presentation. Let me see. I think we still have time. I can.
I can explain, then we will. Let's see where I will reach.
So the system is that the system state is the number of customers in the system. So system state is the number of customers in the system. It is a non-negative integer number. You cannot have a negative number of customers. The number of customers is either.
101 and then it's it's it's an integer. You cannot have a negative number of customers and it's an integer. You cannot have 20.5 number of customers. OK, say that two or three.
And then we let P&T to be the probability that there are N customers in the system at time T So this P&T symbolizes probability that there are N customers in the system.
At time tea.
Then we can present these processes by state transition diagram.
We can we can present it as a state transition diagram. An arrival transforms the system from its current state N to the next state N + 1 any arrival. So if.
If if this is the system, this is a probability that there are N number of customers at time T.
So if there is one more arrival then we have N + 1 customer. So the system then

transformed to state N + Y.
OK, so the system then transforms from state N to state N + 1.
Right. The transition rate from a lower to a higher state, that is what we call arrival is therefore follows, which is therefore what we call Lambda or I think technically we call that.
Intensity of the system, we call it the intensity of the system. So we assume that the rate of arrival is this that's which is typically a Poisson at the rate of that Lambda.
Similarly, as customers are served by a server, the system undergo transition from higher to lower. So once the server takes care of one client, then the system, so the customer has to depart.
As to leave out of the system.
That is referred as the patch our death, so the customers to leave.
The rate at which these departures occur is just the service rate, which is mu, the service mechanism which is mu.
All right, let me before I go there, let me just give us a thought recap or I you see, just trying to see if I can avoid starting the maths, the the main mathematics today and just give us a recap of two.
Mm.
Of two distributions that are of important to us.
One of it we've looked at.
So one distribution we've looked at is Poisson.
Distribution.
So there's a recap of.
Um.
Just a recap of this distribution before you even embed them within the queuing system. So we did show from.
We did show that the poison um.
20.
Will you follow this?
This kind of structure.
My X moves from 01.
To Infinity because we got, if you remember we got, we got, we got this T after doing all the.
After doing all the mathematics.
We ended U with this.

So after after building it from just from the normal from from the Chapman Kolmogorov postulates, we were able to show that eventually we ended up with this type of poison, this type of structure.
And of course now this this one now is like our Lambda. So you can transform that to your Lambda and then N is X. So we end up with with a Poisson.
OK, so the the mean of Poisson.
And using different methods, you can use different methods to get the mean. You can use the what you call the method of moment or you can use the.
Max moment generating function. You can use probability generating function in order to come up with with the the methods with the mean of the Poisson. So the mean of the Poisson depending on the method you use.
You should be able to get Lambda. I think that should be easy for you depending on the method you use if you use MGF.
Where you introduce the EXT and then you rearrange, reorganize, you get that if you use the method of moment. Basically what that that implies is that is is that you open up.
You open up Poison.
OK.
So instead of X factorial here, you could break it into X -, 1 here.
And then if you bring the then this will cancel that and then you can bring this one outside so that you have E so you can have that. So because you cancel this should start from 1:00 so that this is.
Yes, I'm just doing what it's should be easy for you to see. So then if you adjust these, it's just a method of moment. So if you adjust that.
It.
So that this part here gives you a.
Yeah, so that when you multiply this E with E power negative that and E positive that, so you end up with you end up with the Lambda. So that's how we get the mean over over Poisson. OK, the variance should give you the same.
The variance or Poisson just using the same concept. So variance is the second moment subtracted to.
It.
Uh.
OK, so that's that's how you get the variance. Of course we already know this part. Yeah, it's Lambda, so we just need to square it. So this is the second moment and

you can use maximum.
So you can use you can use different methods but using the method of moment.
I can split.
That into this.
OK.
Yeah.
It should be a factorial.
OK, so there's a way you can manipulate, play around with this by breaking this, splitting this factorial and then cancelling and this should be able to give you. I should not even go there because I think you guys know that.
I should be able to give that.
So that the variance now is.
So that the variance now is just Lambda, so the variance and the mean so the the variance.
The mean and is equals to the variance.
And the question of the info is on.
OK, so in Poisson the variance is the same as the mean.
And it's just a and this is what we call that's that's now the Poisson. That's the test for the Poisson.
If if I look at exponential.
So if I look at the exponential distribution.
So for exponential distribution is.
Is um.
Is a continuous function.
X is greater than 0 and then it is 0 as soon.
So that's the exponential distribution. It's a continuous function where X ranges between zero and Infinity. So they mean you can also use different methods, either MGF or you can use.
The method of moment. If you use the MGF, introduce MT and then you partially different get the MGF, then partially differentiate, put it to whichever way you use it.
Let me see.
Does it involves a bit of duration by part? Let me see if. So if you had to get the mean, you have to do 0 to Infinity X mean.
OK, so if you do, if you bring the meal outside.
Just a way integration you can pull out a constant.

And I was pull out in.
So this one require integration by parts. So if you have a product, if you have a product of two functions and you're integrating, if you're differentiating product, we talk of partial, we talk of product rule in differentiation, but integration.
We talk of integration by part. It's just another way of looking at partial integration by part, but looking at it in a different way.
So if you do differentiation by part.
Not going to that.
It's just a technique for differentiation.
To do differentiation by part, you should be able to get one over meal for me and then for variance.
Wait.
It's, you know, the second.
So this one is already known as one and then this one.
To integrate by part again, this is now you have to do twice integration by part.
This one if you do.
I don't know. I think I got lost. Can you hear me now?
I disappeared a bit.
Yeah.

CK **Cosmas Kungu** 2:05:55
Yes, we can hear you. Thank you.

OC **Odhiambo, Collins** 2:05:58
Alright, so if you do again this one differentiation by part, you'll find.
I don't want to go to that because it's it's a bit long. I'll leave it for you to try. So if you do that differentiation by part, you'll have that, which means that variance.
X.
It's equals to E squared.
June.
So then you'll end up with.
Is it say you'll end up with one over MU squared. So I'm just recap recapping the exponential. This is not a probability cost, but this cost assumes.
We have a bit of foundation on probability that the mean of an exponential is given as this and the standard deviation or variance of an exponential is given as that.

If you go to.
If you go to uh
If you go to Poisson, on the other hand, the mean is given as Lambda and the variance is given as.
And you can prove this. This can be proven mathematically. There are many several techniques for showing you can use method of moment like this. This is now the method of moment. Of course there's a step up jumped assuming you did this in undergraduate or at a different form.
And then exponential. Here is where there's a bit of trouble. When you're getting the mean of a mean and variance of an exponential distribution, you required to apply because of the product of two, integrating the product of two.
You need to apply integration by parts because of the product and here you have to apply integration by parts.
Twice. OK, because of the product of two functions. So there's this function and there's that function when it comes to when it comes to Poisson, because Poisson is a discrete distribution.
You just you just play around with the with the function. So you split and then play around with the function. Adjust the function until you get the you get it right.
Right, so.
Um.
The the transition rate. OK, so that's fine. I don't want to introduce them much for this. Let me just see if I can finish with a theory the next week.
I can introduce the maps.
The math should start here.
I don't want to introduce the maths today. Let me just finish the theory part so I could finish this part here.
Then we look at the mathematics next week.
So we are determined or interested in determining the basic measures of performance of a system, such as the probability that N customers are present in the system.
The expected number of customers in the system, the expected number of customers in a queue, the expected number of customers in service, the average waiting time in the system, the average waiting time in the queue.
And the average waiting time in service.
The typical is 4 things LLQLS and then LWQ and WS.

And then given the transition diagram for M1MM2, we can determine the performance measure. A steady state that enters the system should be the same as what comes out. I think I'll I'll explain that so that we can derive that. Let me not introduce that.
Now we can introduce. I can bring in the mathematics next week.
OK.
I'll bring in the entire months. I'll explain the entire months next week so so that we can derive this.
6 steps.

CK **Cosmas Kungu** 2:11:42
And uh, cross before you close.
I note that last week's materials were missing, especially the notes. So if you you'll be uploading for these weeks, also include the last week's notes.

OC **Odhiambo, Collins** 2:11:58
Oh, OK.

CK **Cosmas Kungu** 2:12:00
Yeah.
Especially on those call mother. I tried Googling but now I could not find a good one.
I only found some few. The ones which I found were the one on.
Um.

OC **Odhiambo, Collins** 2:12:16
You know, I I know it's stochastic. You may even find very difficult ones that you can't comprehend, I know.

CK **Cosmas Kungu** 2:12:23
Yeah, and now with with the semester being short, we want to grab the shortest route possible for the escape. Yeah, it's very important for us not to drag along while we we are aiming to graduate next year.

OC **Odhiambo, Collins** 2:12:31

OK, OK, OK.
But.

CK **Cosmas Kungu** 2:12:39
At the same time. So we want the shortest, the lightest basket we can have to carry.
Yeah. Thank you. Thank you.

OC **Odhiambo, Collins** 2:12:43
Yeah.
OK.
OK, OK. OK.
Right. So that that's fine. Let me let me just put them now.
Yeah. So any question, I think we can stop there. So I don't want to introduce the the other mathematics. Once I introduce, we have to move.
With that.
Alright, so you guys enjoy your day then your evening.


Elias Shamalla 2:13:27
OK. Thank you.

SA **Straton Amodora** 2:13:32
Thank you.

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